Bernoulli Types

An Algebra of Approximate Computation

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ORCID: `0000-0001-6443-9897`
Web: `https://metafunctor.com`
Source: `https://github.com/queelius/bernoulli-book`

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Preface

This is a placeholder preface. The final preface will:

- State the book's thesis and audience.
- Declare prerequisites: undergraduate probability, basic discrete math, comfort with proof, working Python knowledge for the notebook companion.
- Explain the lockstep notebook discipline (some chapters carry notebooks; theory-heavy chapters do not, by design).
- Provide a reading guide for the smart-undergrad-upward reader (chapters that admit a soft landing) and the adjacent-field grad student (the modal reader).

Part I

The Atom

Chapter 1

The Noisy Bit

Placeholder. Content drafted in Plan 1. The framework appears in [Tow26a].

Chapter 2

Binary Channels and Model Order

Placeholder. Content drafted in Plan 1.

Chapter 3

Boolean Algebra Under Noise

Placeholder. Content drafted in Plan 1.

Chapter 4

Two Axioms and Kronecker Factorization

After this chapter you will be able to:

- State Axioms 1 and 2 of the Bernoulli model formally.
- Prove the Kronecker factorization theorem and derive the parameter count drop from $n(2^m - 1)$ to $2m$.
- Write the formal definition of Bernoulli[bool] and recognize every example from chapters 1–3 as an instance.

4.1 Why Axioms?

The three chapters behind us built something real. We derived the error rates for AND and OR gates operating on noisy bits (**??**), showed that composition of channels produces predictable compound rates, and in **??** proved the composition theorem: when k independent noisy channels are chained in series, their true-positive rates compose multiplicatively. We then applied that theorem immediately to Miller–Rabin primality testing, watching a per-round false-positive rate of at most $1/4$ collapse to $(1/4)^k$ after k rounds. The derivations were concrete and the results were correct.

But correct derivations can hide assumptions. Chapters 1–3 made *two* independence assumptions so naturally that we did not pause to name them. This chapter names them, states them precisely, and then shows exactly what mathematical structure they license. That structure is a precise structural theorem showing what those assumptions cost and what they buy; it is the formal payoff for the care we are about to take.

The first hidden assumption: element-wise independence

Recall the composition theorem from **??**. When we wrote the true-negative rate for k channels chained in series as $\eta_{\text{total}} = 1 - \prod_{i=1}^{k}(1 - \eta_i)$, we were treating each channel's contribution to the end-to-end error as independent of every other channel's contribution. One query enters the chain; it passes through channel 1, whose output feeds channel 2, and so on; the overall error rate is a product of independent per-stage terms. No correlation across stages was allowed, and the product formula relied on that fact throughout.

A different scenario where Axiom 1 matters: two classifiers trained on overlapping data from the same source. Their errors on a new input x will tend to correlate: when one misclassifies x, the other is more likely to as well. The product formula breaks. The effective error rate is worse than the product formula predicts, because the errors reinforce rather than cancel.

We will call the assumption that justifies the product formula *Axiom 1*. Roughly, Axiom 1 asserts independence of error events across distinct input elements; the precise statement and its scope appear in Section 4.2. The practical effect is that joint error distributions factor across distinct queries, which is what made the composition theorem's product form possible. When Axiom 1 holds, every formula in chapters 1–3 that relies on independence is automatically justified. When Axiom 1 fails, those formulas become bounds at best and misleading at worst. Chapter 3's Miller–Rabin analysis, for instance, requires Axiom 1: the k witness tests must be independent for the per-round bound to compound as $(1/4)^k$. If the witnesses were correlated, the product would underestimate the true composite error rate.

The second hidden assumption: stable block error rates

The derivation of AND under noise in **??** used specific values ε and ω, the false-positive and false-negative rates, treating them as fixed properties of each input. We implicitly assumed that querying an input repeatedly would yield the same error behavior each time: that ε does not drift from query to query, that the channel is stationary across uses.

This is not the only conceivable behavior. A Markov-correlated channel, for instance, lets the error rate on query $t + 1$ depend on what happened at query t. If the channel just made a false positive, it might be more likely to make another. The rates are not stable; they are history-dependent. We did not model this, and for good reason: the stationary case is the tractable and practically important one. But we should say we are assuming it.

We have been assuming Axiom 2 since **??**; let us make it explicit. Axiom

2 says: the error rates within each partition block of the input domain are the same for every query, unconditionally. There is one channel matrix Q per input-block-pair, and it does not depend on which query in a sequence we are currently processing. Under Axiom 2, the channel is genuinely memoryless within each block: its statistical behavior is the same on the hundredth query as on the first.

This is a meaningful restriction, not a triviality. It rules out fatigue, warm-up effects, adversarial adaptation, and history dependence. It also makes the math work cleanly. We formalize Axiom 2 in Section 4.3.

What the axioms buy

Once both axioms are in place, something striking follows. The joint confusion matrix describing m independent queries on the channel is not a free matrix with exponentially many parameters. Under Axioms 1 and 2, it *must* factor into a product of m small per-query matrices, one factor per query. The exponential-to-linear collapse in parameter count, made precise in Section 4.4, is the formal payoff of this chapter.

This is not merely an algebraic convenience. It is the theorem that makes the Bernoulli framework useful in practice. Without it, a fifty-query analysis would require specifying an astronomically large matrix. With it, you need fifty pairs of numbers. The saving is exponential.

We prove this Kronecker factorization theorem, derive the parameter count corollary, and work through explicit examples in Section 4.4. The theorem is the chapter's climax and the formal vindication of every computation in chapters 1–3.

The formal definition

Once the axioms and the Kronecker theorem are in hand, we can at last write down the formal definition of the type Bernoulli[bool] without fear of circularity. Chapter 1 named the noisy bit informally. Chapter 2 gave it a channel matrix. Chapter 3 showed how to compose these objects. Chapter 4 closes by showing that all of those things are instances of a single precise definition: a triple (X, Y, Q) where X and Y are the input and output domains $\{\top, \bot\}$, Q is a 2×2 stochastic matrix, and the whole structure satisfies Axioms 1 and 2. We work this out in Section 4.5, including the biased-coin example from §1.3 that was deliberately left without a full PMF until we had the formal machinery to handle it.

The trajectory of this chapter, then, is: axioms first (Sections 4.2 and 4.3),

then the theorem they make possible (Section 4.4), then the definition the theorem earns us (Section 4.5).

By the end, the reader who worked through chapters 1–3 will recognize every example, every formula, and every derivation as an instance of one coherent definition. That is the point.

4.2 Axiom 1: Element-Wise Independence

The previous section sketched two kinds of independence that chapters 1–3 quietly assumed. This section makes the first of them precise.

A brief note on framing before we begin. In Section 4.1 we used a chain-of-stages example as motivation: k noisy channels in series, with end-to-end error rate $1-\prod_{i=1}^{k}(1-\eta_i)$. That example invokes a kind of independence, but it is not the same statement as Axiom 1. Axiom 1 governs independence across distinct input elements of a *single* approximate predicate. Stage-wise independence in a composition chain is a separate assumption: closely related, often co-occurring, but operationally distinct. Both are needed for the chain formula; both fail in similar circumstances; and in well-behaved implementations both hold. We state Axiom 1 cleanly here, on a single predicate. The reader who wants to recover the chain formula from these axioms should track the two independences separately and combine them as the composition theorem of **??** prescribes.

The formal statement

Let U be the universal set, the domain of all possible inputs to the approximate membership predicate. For a fixed objective set $A \subseteq U$, the predicate either correctly reports the true answer or makes an error. For each $x \in U$, define the error indicator

$$E_x \;=\; \mathbf{1}\big[\tilde{A}(x) \neq \mathbf{1}_A(x)\big],$$

where $\tilde{A}(x)$ is the approximate predicate's output on input x and $\mathbf{1}_A(x)$ is the exact answer. $E_x = 1$ means an error occurred at x; $E_x = 0$ means the answer was correct.

Axiom 1 (Element-wise independence)**.** *For all distinct $x, y \in U$, the error events E_x and E_y are independent:*

$$\mathrm{P}\left(E_x = 1,\, E_y = 1\right) \;=\; \mathrm{P}\left(E_x = 1\right) \cdot \mathrm{P}\left(E_y = 1\right).$$

More generally, the family $\{E_x : x \in U\}$ is mutually independent given the objective set A and the error rates. Furthermore, within each partition block B_i of U (defined formally in Section 4.3), the error indicators $\{E_x : x \in B_i\}$ are identically distributed.

The notation $\tilde{A}(x)$ is deliberately abstract: it stands for any approximate membership test, whether implemented as a Bloom filter, a probabilistic data structure, a noisy sensor, or a randomized algorithm. Axiom 1 is a property of the *process* that generates the answers, not of any particular implementation.

Consequence: factored joint distributions

Axiom 1 is not merely a philosophical assertion. It has an immediate and quantitative consequence for joint query distributions.

Proposition 4.2.1 (Factored joint distribution)**.** *Under Axiom 1, the joint distribution of the error indicators over any finite set of distinct inputs $\{x_1, \ldots, x_n\} \subseteq U$ factors as a product of n marginal distributions:*

$$\mathrm{P}\left(E_{x_1} = e_1,\, \ldots,\, E_{x_n} = e_n\right) \;=\; \prod_{i=1}^{n} \mathrm{P}\left(E_{x_i} = e_i\right)$$

for any choice of $e_1, \ldots, e_n \in \{0, 1\}$. Equivalently, the joint distribution of the observed responses $\tilde{A}(x_1), \ldots, \tilde{A}(x_n)$ factors as

$$\mathrm{P}\left(\tilde{A}(x_1) = s_1, \ldots, \tilde{A}(x_n) = s_n | A\right) \;=\; \prod_{i=1}^{n} \mathrm{P}\left(\tilde{A}(x_i) = s_i | A\right),$$

where each factor equals ε, η, τ, or ω depending on whether $x_i \in A$ and whether s_i is a correct response.

Proof. The first identity is the definition of mutual independence for the family $\{E_x : x \in U\}$, which Axiom 1 asserts directly. The second identity follows because $\tilde{A}(x_i) = s_i$ is a deterministic function of E_{x_i} and the exact answer $\mathbf{1}_A(x_i)$: knowing A determines $\mathbf{1}_A(x_i)$, and given that, $\tilde{A}(x_i) = s_i$ iff E_{x_i} takes the appropriate value. The product therefore inherits its factored form from the independence of $\{E_{x_i}\}$. □

This proposition is the engine behind every multi-query formula in the book. Whenever we write a joint probability over several distinct inputs as

a product of marginals, we are using this proposition, which rests on Axiom 1.

The factored joint distribution also licenses a key observation about the joint confusion matrix of m simultaneous queries. Without any independence assumption, that matrix lives in a space of dimension $n(2^m - 1)$ for an n-th order model. Axiom 1 forces the rate-conditional joint matrix to factor as a Kronecker product of per-query 2×2 matrices, collapsing the parameter count to at most $2m$. That is the content of Section 4.4; we state it here as a forward reference so the reader knows why Axiom 1 matters at the level of the whole framework.

Example 1: The Bloom filter's independence assumption

A Bloom filter [Blo70] answers membership queries on a set A stored in a bit array of m bits, using k independent hash functions. When k hashes are applied to a new element x, each hash selects a bit position; x is declared a member if and only if all k selected positions are 1.

For elements $x \notin A$, the probability that all k positions happen to be set (a false positive) is approximately $(1 - e^{-kn/m})^k$, where $n = |A|$. This familiar formula is derived under an explicit assumption: for two distinct non-members x and y, the event "x is a false positive" is independent of the event "y is a false positive." Without this assumption, the formula gives neither an exact probability nor a useful bound; instead, one would need to account for correlations between all pairs of non-members, producing a quantity with no closed form.

Axiom 1 is exactly the assumption the Bloom filter's analysis needs. The filter's k hash functions are chosen to be independent, and the hash values for distinct elements are treated as unrelated. Whether or not this is precisely true in a given implementation, it is the assumption under which the false-positive rate formula is derived, and it is the assumption that the Bernoulli framework formalizes.

We will analyze the Bloom filter fully in Part II (Chapter 5), where Axiom 1 will be verified to hold under the standard random-oracle model for the hash functions. For now, the take-away is structural: the Bloom filter is a Bernoulli[bool] implementation, and Axiom 1 is the assumption that makes that claim non-trivial.

Example 2: Miller-Rabin's k-round bound requires Axiom 1

In **??** we derived the Miller–Rabin bound: after k independent rounds of witness testing, the probability that a composite number passes all k rounds (the k-round false-positive rate) is at most $(1/4)^k$. The bound rests on two facts: each round's false-positive probability is at most $1/4$, and the rounds are independent.

There is a subtle reframing required to fit Miller-Rabin under Axiom 1. The predicate being tested takes a *single* fixed input n (the integer in question), and randomness enters via the witness draw rather than via the input. To apply Axiom 1 cleanly, recast the test as a predicate $\tilde{P}\colon \{2,\dots,n-2\} \to \{\text{composite}, \text{probably prime}\}$ on the witness pool: now distinct witnesses *are* distinct input elements, and Axiom 1's claim is that error events at distinct witnesses (the events "w_i is a strong liar for n") are independent. Whether or not this independence holds for any particular composite n depends on the structure of n's strong-liar set, a separate fact about Miller-Rabin's mathematical content.

With that reframing in place, the connection to Axiom 1 is clean. Each round tests a different witness w_i drawn uniformly at random from $\{2,\dots,n-2\}$. Whether w_i happens to be a strong liar for n (the error event at w_i) must be independent of whether w_j is a strong liar (the error event at w_j), for $i \neq j$. The error events are the events that the witness fails to detect compositeness.

Under Axiom 1, the joint probability that all k witnesses are strong liars is the product of the individual probabilities:

$$\mathrm{P}\,(\text{all } k \text{ witnesses are strong liars}) \;\leq\; \prod_{i=1}^{k} \frac{1}{4} \;=\; \left(\frac{1}{4}\right)^{k}.$$

Without Axiom 1, the k error events could be positively correlated. The events "w_1 is a liar" and "w_2 is a liar" may be positively correlated, because both witnesses are sampled from the same composite n, which may itself happen to have an unusually large strong-liar set. Conditioning on n, the witnesses are not independent of each other when the liar fraction is unknown. In that case, the product bound $(1/4)^k$ would underestimate the true false-positive probability. The correct expression would involve the joint distribution over all k witnesses, which does not simplify to a product without an independence assumption.

The Miller–Rabin analysis works because the witnesses are drawn uniformly and independently from a large set. In the worst case, at most $1/4$

of all candidates are strong liars; with independent uniform sampling, the fraction is driven down exponentially. This is Axiom 1 at work.

Classifier-verdict callback: a related independence failure

The classifier-verdict thread from Section 4.1 returns here, but with a deliberate caveat. Two correlated classifiers evaluating the *same* input is not literally an Axiom 1 failure: Axiom 1 governs distinct *inputs* to a single predicate, not distinct *predicates* on a single input. The example we work below is one step removed from Axiom 1, but it illustrates the same family of phenomenon: when error events are correlated for any reason, the naive product formula underestimates the joint failure rate. The Axiom 1 case (correlated errors at distinct inputs to one predicate) is mathematically structurally identical.

Suppose two classifiers C_1 and C_2 are both trained on a shared dataset with overlapping features. Each classifier is a Bernoulli[bool] individually: on any given input x, C_1 has false-positive rate ε_1 and C_2 has false-positive rate ε_2.

Now consider their joint behavior on a single negative input $x \notin A$. If the errors were independent, the probability that both make a false positive would be:

$$\mathrm{P}\left(C_1(x) = 1,\, C_2(x) = 1 | x \notin A\right) \stackrel{?}{=} \varepsilon_1 \cdot \varepsilon_2.$$

But the two classifiers share training data, so their errors on x tend to be correlated.

Let $\rho \in [0, 1]$ be the correlation between the error indicators $E_x^{(1)}$ and $E_x^{(2)}$. The joint probability is

$$\mathrm{P}\left(E_x^{(1)} = 1,\, E_x^{(2)} = 1\right) \;=\; \varepsilon_1\varepsilon_2 \;+\; \rho\sqrt{\varepsilon_1(1-\varepsilon_1)\cdot\varepsilon_2(1-\varepsilon_2)}.$$

For a concrete example, let $\varepsilon_1 = \varepsilon_2 = 0.10$ and $\rho = 0.60$. The product bound predicts a joint false-positive rate of $0.10 \times 0.10 = 0.01$. The actual joint rate is

$$0.01 + 0.60 \times \sqrt{0.10 \times 0.90 \times 0.10 \times 0.90} \;=\; 0.01 + 0.60 \times 0.09 \;=\; 0.01 + 0.054 \;=\; 0.064.$$

The inflation factor is $0.064/0.01 = 6.4$: the true joint false-positive rate is $6.4\times$ the naive product.

This is not a failure of the individual classifiers, which are each well-behaved Bernoulli[bool] objects; it is a failure of the independence assumption between them. The Axiom 1 violation has the same arithmetic; only the source of the correlation differs.

The twin lessons are these. First, Axiom 1 is a genuine modeling assumption, not an automatic consequence of each individual predicate being a Bernoulli[bool]: the individual marginals can be well calibrated while the joint distribution fails to factor. Second, correlated errors are worse than independent errors in a precise, quantifiable sense: the inflation factor grows with the correlation coefficient and with the individual error rates.

The Kronecker theorem proved in Section 4.4 makes this precise at the level of the joint confusion matrix: under Axiom 1, the matrix has a structured Kronecker-product form that encodes exactly the factored joint distribution in Proposition 4.2.1. Without Axiom 1, the matrix is unstructured, the parameter count explodes, and the elegant formulas of chapters 1–3 must be replaced by unwieldy joint distributions over all pairs and triples of query events.

4.3 Axiom 2: Conditional Independence of Block Error Rates

Section 4.2 gave the first independence axiom: error events at distinct inputs are mutually independent, given the objective set and the error rates. That axiom governs the *spatial* structure of errors, the relationship between different elements of the universal set queried simultaneously or in any order.

This section gives the second axiom. Axiom 2 governs how the random error rates at different blocks relate to each other: knowing the rate on one block tells you nothing about the rate on another. This is a *cross-block independence* claim, not a within-block or across-queries stability claim. Combined with Axiom 1, however, it implies a stability we will derive as a formal consequence in Proposition 4.3.1: per-block rates are constant across queries. We state the formal axiom first, then derive the stability consequence, then work through what fails when stability fails.

The two axioms are logically independent of each other. Axiom 1 alone does not imply Axiom 2, and Axiom 2 alone does not imply Axiom 1. Both are needed, and their conjunction is exactly what makes the Kronecker factorization theorem of Section 4.4 possible.

Partition blocks and model order

Before stating Axiom 2, we need the partition structure it ranges over. A full treatment appears in **??**; we summarize here the minimum needed for the axiom statement.

Given an objective set $A \subseteq U$, the Bernoulli model partitions the universal set U into blocks $B_1, \ldots, B_n$ such that every element within a block has the same error rate. The *model order* is the number of blocks n. The simplest nontrivial case is order 2: the partition $\{B_1, B_2\} = \{A,\ A^c\}$ separates the positives from the negatives and assigns each group its own error rate.

For the Boolean universe $U = \{\top, \bot\}$ with objective set $A = \{\top\}$, model order 2 yields two blocks:

$$B_1 = \{\top\} = A,$$
$$B_2 = \{\bot\} = A^c.$$

Elements of B_1 have false-negative rate ω (they test positive correctly with probability τ, incorrectly with probability ω). Elements of B_2 have false-positive rate ε (they test negative correctly with probability η, incorrectly with probability ε). The channel matrix Q introduced in **??** captures exactly these two rates:

$$Q \;=\; \begin{pmatrix} \tau & \omega \\ \varepsilon & \eta \end{pmatrix},$$

where row i describes the output distribution for an element in block B_i.

In general, an n-th order model has n blocks and a corresponding channel matrix with n distinct rows, one per block. Axiom 2 is a statement about those rows.

The formal statement

Axiom 2 (Conditional independence of block error rates)**.** *Let $\{\alpha_1, \ldots, \alpha_n\}$ be the random error rates for the respective partition blocks $B_1, \ldots, B_n$, conditional on the objective set A. The rates $\alpha_1, \ldots, \alpha_n$ are mutually independent:*

$$f(\alpha_1, \ldots, \alpha_n | A) \;=\; \prod_{i=1}^{n} f(\alpha_i | A).$$

In plain language: knowing the error rate on positives tells you nothing about the error rate on negatives, and vice versa. The block rates are independent of each other given the objective set. This applies to all n blocks simultaneously: no block's rate gives information about any other block's rate.

A few clarifications help prevent common misreadings.

Rates as random variables. The rates α_i are random in the formal sense: they are the *expected* error frequencies, which may themselves vary

across different realizations of the data structure or algorithm. A Bloom filter, for instance, has a false-positive rate that depends on the specific set of inserted elements and the hash function choices. Axiom 2 says that the resulting FPR and FNR are independent across blocks.

Stability, not constancy. Axiom 2 does not say the rates are the same for every element within a block (that is the identically-distributed clause of Axiom 1). It says the rates are jointly independent across blocks. The distinction matters: a system could have rates that vary from element to element within a block while still satisfying Axiom 2. The axioms play complementary roles; neither subsumes the other.

The order-2 specialization. In the second-order model, Axiom 2 reduces to a single statement: the random false-positive rate ε and the random false-negative rate ω are conditionally independent given A. The joint density of the two rates factors:

$$f(\varepsilon, \omega | A) \;=\; f(\varepsilon | A) \cdot f(\omega | A).$$

This is the instance that arises in virtually every example in chapters 1–3.

Consequence: one channel matrix per block, stable across queries

Axiom 2 has a structural consequence for how we model repeated queries. Suppose we query an input $x \in B_i$ multiple times: first at step t, then again at step t', and so on. Axiom 2 says the error rate on query at step t does not influence the error rate on query at step t'. Combined with Axiom 1 (which says errors at distinct elements are independent), this means:

Proposition 4.3.1 (Stable channel matrices)**.** *Under Axiom 2, the channel matrix row corresponding to block B_i is the same for every query involving an element of B_i, regardless of the order or history of prior queries. Formally, for any sequence of queries $(x_1, x_2, \ldots)$ with $x_t \in B_{i(t)}$, the conditional distribution*

$$\mathrm{P}\left(\tilde{A}(x_t) = s | A,\; x_t \in B_{i(t)}\right)$$

is the same function of $i(t)$ and s for all t, independent of what happened at steps $1, \ldots, t-1$.

Proof. Fix block B_i and a sequence of queries. The distribution of the output at step t for $x_t \in B_i$ is determined by the rate α_i assigned to block B_i. By Axiom 2, α_i is independent of α_j for $j \neq i$. By Axiom 1, the error events at

distinct elements are independent. Together, the conditional distribution of the output at step t is a function only of α_i and the block membership of x_t, not of any prior query outcomes.

By Axiom 1, every element of B_i uses the same rate α_i, drawn once at construction time for the data structure. Once that value is realized, the channel row for B_i is a fixed function of α_i at every subsequent query, regardless of the query index t. Marginalizing over the distribution of α_i then produces the same marginal channel row at every query. Because Axiom 1 also makes per-element error events conditionally independent given α_i, every query of an element from block B_i contributes the same row of Q. The Q matrix is therefore time-invariant across the query sequence. Since this holds for all t, the channel matrix row for block B_i is stable across the entire query sequence. □

This proposition is the precise sense in which "there is one channel matrix Q per input-block-pair." The matrix entry Q_{ij} gives the probability that an element of block B_i produces output j. That probability does not drift, does not warm up, and does not depend on how many prior queries have been processed. The channel is genuinely stationary within each block.

The practical consequence is large. Without this stability, we would need a separate channel description for each time step, and the joint distribution over a sequence of queries would require specifying an exponentially growing family of time-varying matrices. With Axiom 2, a single matrix Q suffices for the entire query sequence. This is what makes the Kronecker factorization of Section 4.4 possible: the m-query joint confusion matrix is the Kronecker product of m copies of the same Q, precisely because each query sees the same stable channel.

Worked illustration: the model-order-2 Boolean channel

Let us make Axiom 2's content concrete with the simplest nontrivial case, which will recur throughout the book: the Boolean universe with model order 2.

Set $U = \{\top, \bot\}$ and $A = \{\top\}$, giving two blocks: $B_1 = \{\top\}$ (positives) and $B_2 = \{\bot\}$ (negatives). The channel matrix is

$$Q = \begin{pmatrix} \tau & \omega \\ \varepsilon & \eta \end{pmatrix} = \begin{pmatrix} 1-\omega & \omega \\ \varepsilon & 1-\varepsilon \end{pmatrix}.$$

Now consider a sequence of three queries: query 1 asks for $\top$, query 2 asks for $\bot$, query 3 asks for $\top$ again. Axiom 2 says the following.

- Query 1 involves $\top \in B_1$. It produces a positive response with probability τ and a negative response with probability ω. These probabilities come from row 1 of Q.

- Query 2 involves $\bot \in B_2$. It produces a positive response with probability ε and a negative response with probability η. These probabilities come from row 2 of Q. *The outcome of query 1 is irrelevant.*

- Query 3 involves $\top \in B_1$ again. It again uses row 1 of Q, with the *same* probabilities τ and ω as query 1. The fact that query 1 may have made an error (false negative) does not change the probability of error on query 3.

This is the channel's stationarity within blocks: the same row of Q applies whenever the same block is queried, no matter what came before.

To write the joint distribution over the three-query sequence $(x_1, x_2, x_3) = (\top, \bot, \top)$ with outcomes (s_1, s_2, s_3), Axioms 1 and 2 together give:

$$\begin{aligned} &\mathrm{P}\left(\tilde{A}(\top) = s_1,\ \tilde{A}(\bot) = s_2,\ \tilde{A}(\top) = s_3 | A\right) \\ &\quad = \ Q_{1,s_1} \cdot Q_{2,s_2} \cdot Q_{1,s_3}, \end{aligned}$$

where $Q_{i,s}$ denotes the (i, s)-entry of Q. The distribution factors as a product of independent channel evaluations, each determined by the block of the queried element. No interaction terms appear; no history from query 1 flows into query 3.

For a concrete numerical example, take $\varepsilon = 0.05$ and $\tau = 0.90$ (so $\omega = 0.10$ and $\eta = 0.95$). The probability that the three-query sequence produces outcomes $(+, -, +)$, that is, two true positives and one true negative, is:

$$Q_{1,+} \cdot Q_{2,-} \cdot Q_{1,+} \ = \ 0.90 \times 0.95 \times 0.90 \ = \ 0.7695.$$

The probability that query 2 produces a false positive while queries 1 and 3 produce true positives is:

$$Q_{1,+} \cdot Q_{2,+} \cdot Q_{1,+} \ = \ 0.90 \times 0.05 \times 0.90 \ = \ 0.0405.$$

Both numbers follow from reading the appropriate rows of Q and multiplying. No additional modeling is needed. The stationarity guaranteed by Axiom 2 is what makes this reading-and-multiplying procedure valid.

Contrast: a Markov-correlated channel violates Axiom 2

To see what Axiom 2 rules out, consider a channel that violates it explicitly. A *Markov-correlated channel* lets the error rate on each query depend on the outcome of the previous query. Concretely:

1. If the previous query produced a correct response, the current query uses a "base" false-positive rate ε_0.

2. If the previous query produced an error, the current query uses an elevated false-positive rate $\varepsilon_1 > \varepsilon_0$ (errors cluster together).

This is a plausible model for hardware memory with correlated faults, or for a sensor that drifts after one bad reading. But it violates Axiom 2.

To see why, consider block $B_2 = \{\perp\}$ (negatives). On query t, the error rate for block B_2 is not a fixed value ε: it is ε_0 if the previous query was correct, and ε_1 if the previous query was an error. The rate α_2 at step t is not independent of the rate at step $t-1$; they are coupled through the Markov chain. The joint density of the rates at different steps does not factor, and Axiom 2's independence condition fails.

Let us put numbers to this. Suppose the base rate is $\varepsilon_0 = 0.05$ and the elevated rate is $\varepsilon_1 = 0.30$. The channel starts in the correct state. Over two successive queries to $\perp$, the joint distribution of outcomes is:

Outcome pair	Probability	Axiom-2 product	Discrepancy
$(-,-)$	$0.95 \times 0.95 = 0.9025$	0.9025	none
$(-,+)$	$0.95 \times 0.05 = 0.0475$	0.0475	none
$(+,+)$	$0.05 \times 0.30 = 0.0150$	0.0025	$6\times$ inflation
$(+,-)$	$0.05 \times 0.70 = 0.0350$	0.0475	deflation

Axiom 2 predicts the joint probability of two consecutive false positives as $\varepsilon_0^2 = 0.05^2 = 0.0025$, since under Axiom 2 the per-block false-positive rate is a fixed constant ε_0 across queries (one Q matrix per block, stable). The Markov channel produces 0.0150 instead, a factor of $6\times$ higher. The discrepancy is exactly what Axiom 2 forbids: the rate after a previous false positive ($\varepsilon_1 = 0.30$) is six times the rate after a previous correct response ($\varepsilon_0 = 0.05$). The per-block rate is no longer a constant; it depends on history.

The discrepancy is not small. The Markov channel produces error clustering that is invisible to the factored model. If a practitioner were to apply the Kronecker factorization of Section 4.4 to this channel, they would underestimate the probability of back-to-back errors by a factor of six. In

applications where error clustering is costly, this is not a tolerable approximation error.

The Markov channel is not a pathological construction: it models a real failure mode. What Axiom 2 asserts is that the Bernoulli framework, in its standard form, is designed for channels where this clustering does not occur. If the channel is known to be Markov-correlated, the model must be extended, for example by conditioning on the hidden state, or by treating the full trajectory as the object to be analyzed. The Bernoulli axioms are then either violated, requiring a richer model, or satisfied at the level of the extended state space, in which case the standard machinery applies after the appropriate state augmentation.

What Axiom 2 earns us

Axiom 1 and Axiom 2 are the full axiomatic foundation of the Bernoulli framework. They are minimal in the sense that neither alone is sufficient for the Kronecker factorization theorem: Axiom 1 gives spatial independence across elements, Axiom 2 gives temporal stability of rates across queries. Together they imply that the joint confusion matrix over any sequence of queries is a Kronecker product of single-query matrices, one factor per query (Section 4.4).

The parametric savings are substantial. Consider the order-2 Boolean predicate that runs throughout this chapter: a general channel on m joint queries with two blocks, and no independence assumption, requires $2(2^m - 1)$ free parameters: one for each joint outcome of m Boolean tests in each of the two rows, save normalization. Under Axioms 1 and 2, the Kronecker structure forces the parameter count down to $2m$: two parameters per query (one per block), with no cross-query parameters at all. For $m = 50$ queries on the order-2 Boolean channel, that is $2m = 100$ parameters under the axioms versus $2(2^{50} - 1) \approx 2.25 \times 10^{15}$ in the general case. The axioms are not merely philosophically satisfying; they are what makes the model tractable.

One refinement is worth flagging now and pursuing in Section 4.4. The Kronecker structure forces the parameter count down to nm for the rate-conditional channel; the unconditional matrix achieves the same count when no block is queried twice (Corollary 4.4.1.1) or when rates are deterministic (the regime of all worked examples in this book). When two queries hit the same block under genuinely random rates, the unconditional joint involves higher moments of those rates and the simple per-query count no longer applies; Section 4.4 treats this distinction explicitly.

The Kronecker factorization theorem in Section 4.4 makes this precise and derives the parameter reduction as a corollary. For an n-th order model on m joint queries, the general case has $n(2^m - 1)$ free parameters; under the axioms, the count falls to nm.

We prove the Kronecker factorization theorem and its corollary on parameter reduction in Section 4.4. That section is where the full benefit of both axioms becomes visible.

4.4 The Kronecker Factorization Theorem

Two axioms, one payoff

We have named two axioms and shown what each rules out. Section 4.2 gave Axiom 1: error events at distinct inputs are mutually independent. Section 4.3 gave Axiom 2: per-block error rates are conditionally independent of each other, which implies that each block's channel row is stable across queries. Both sections foreshadowed a theorem that their conjunction enables. We now prove it.

The question is this. Given m queries on a Bernoulli predicate, each on a distinct element of the universal set—what does the joint confusion matrix look like? The joint matrix $Q^{(m)}$ is an $n^m \times 2^m$ object: rows are indexed by source-block tuples $(j_1, \ldots, j_m) \in \{1, \ldots, n\}^m$ (one row per ordered selection of source blocks across the m queries), and columns are indexed by binary output patterns $(k_1, \ldots, k_m) \in \{0, 1\}^m$. Without any structural assumption, specifying this matrix requires $n^m(2^m - 1)$ real parameters (one distribution over 2^m output patterns per source-block tuple). For $m = 50$ and $n = 2$, that is over 10^{30} free parameters.

A different, natural parameterization treats only the n distinct source blocks, not all n^m block-tuples, as the basic unit. Under this per-source-block convention the joint conditional distribution over 2^m output patterns is the same for every query from block B_j, giving n distributions of size 2^m and hence $n(2^m - 1)$ free parameters. The parameter counts in this section's corollary and in Section 4.3 are stated under this convention—and it is exactly the convention the axioms enforce, as the corollary explains.

Under Axioms 1 and 2, the per-source-block parameter count drops to nm: n free parameters per query (one per block row), with no cross-query interaction, summed over m queries. In the order-2 Boolean case this is $2m$. This is not an approximation; it is the exact structure forced by independence.

Notation. *The* Kronecker product *of two matrices P and Q, written $P \otimes Q$, is the block matrix formed by replacing each entry P_{ij} with the scaled submatrix $P_{ij}Q$. If P is $r \times s$ and Q is $p \times q$, then $P \otimes Q$ is $(rp) \times (sq)$. The Kronecker product is associative, so $P \otimes Q \otimes R$ is unambiguous without parentheses. We use $\otimes$ for the Kronecker product throughout this section and the rest of the book.*

Theorem 4.4.1 (Kronecker Factorization, conditional form)**.** *Let $\tilde{A}$ be an n-th order Bernoulli predicate satisfying Axiom 1 (Axiom 1) and Axiom 2 (Axiom 2). For any realization of the per-block rates $\alpha = (\alpha_1, \ldots, \alpha_n)$, let $Q_1^*(\alpha), \ldots, Q_m^*(\alpha)$ be the resulting per-query channel matrices, where $Q_i^*(\alpha)$ is the $n \times 2$ stochastic matrix whose (j,k)-entry is the probability, under the realized rate α_j, that a query of an element in block B_j produces output label k on the i-th query. Then the* rate-conditional *joint confusion matrix $Q^{(m)*}(\alpha)$ of m queries on m distinct input elements, whose $(j_1 \cdots j_m,\, k_1 \cdots k_m)$-entry is*

$$\mathrm{P}\left(\tilde{A}(x_1) = k_1, \ldots, \tilde{A}(x_m) = k_m | \alpha,\ x_\ell \in B_{j_\ell} \text{ for each } \ell\right),$$

factors as the Kronecker product

$$Q^{(m)*}(\alpha) \;=\; Q_1^*(\alpha) \otimes Q_2^*(\alpha) \otimes \cdots \otimes Q_m^*(\alpha).$$

Corollary 4.4.1.1 (Unconditional factorization, distinct-block queries)**.** *Under the hypotheses of Theorem 4.4.1, if the queried blocks $j_1, \ldots, j_m$ are pairwise distinct (each query hits a different block), the unconditional joint confusion matrix factors as*

$$Q^{(m)} \;=\; Q_1 \otimes Q_2 \otimes \cdots \otimes Q_m,$$

where $Q_\ell \coloneqq \mathrm{E}\left[Q_\ell^*(\alpha)\right]$ is the rate-marginalized per-query channel.

Before the proof, two notes on scope. **First**, the theorem applies to m queries on m *distinct* input elements. Repeated queries of the same element introduce correlations that Axioms 1 and 2 do not address: the two error events share a single realization of the element's underlying error draw, making them perfectly correlated (in a deterministic implementation) rather than independent. For repeated-query behavior, including the effects of memoization and caching, see Chapter 16 (negative space).

Second, the theorem's primary claim is at the rate-conditional level: given any realization of the per-block rates α, the joint confusion matrix factors. The unconditional matrix factors only when the queried blocks are pairwise distinct (Corollary 4.4.1.1). When two queries hit the same block

B_j, both factors involve the same random rate α_j, and the unconditional joint $\mathrm{E}\left[\alpha_j^2\right]$ exceeds the Kronecker prediction $(\mathrm{E}[\alpha_j])^2$ whenever α_j is non-degenerate (a Jensen gap). The conditional factorization holds always; the unconditional factorization holds when either the rates are realized as constants (the deterministic-implementation case used throughout chapters 1–3) or when no block is queried twice. In the running BSC thread below and in subsequent chapters, we work in the deterministic-rates regime, where $Q_\ell^* = Q_\ell$ and the conditional and unconditional forms coincide.

Proof. We build up the joint distribution by hand and read off the matrix structure.

Step 1: reduce to a joint probability calculation. Index the m query positions as $\ell = 1, \ldots, m$. Position ℓ queries element $x_\ell \in B_{j_\ell}$ and observes output k_ℓ. Fix a realization of the per-block rates $\alpha = (\alpha_1, \ldots, \alpha_n)$. The $(j_1 \cdots j_m, k_1 \cdots k_m)$-entry of the rate-conditional matrix $Q^{(m)*}(\alpha)$ is

$$Q^{(m)*}(\alpha)_{j_1 \cdots j_m,\, k_1 \cdots k_m} \;=\; \mathrm{P}\left(\tilde{A}(x_1) = k_1, \ldots, \tilde{A}(x_m) = k_m | \alpha,\; x_\ell \in B_{j_\ell},\; \ell = 1, \ldots, m\right). \tag{a}$$

We want to evaluate this probability and show it equals $\prod_{\ell=1}^{m} (Q_\ell^*(\alpha))_{j_\ell, k_\ell}$, which is precisely the $(j_1 \cdots j_m, k_1 \cdots k_m)$-entry of $Q_1^*(\alpha) \otimes \cdots \otimes Q_m^*(\alpha)$.

Step 2: apply Axiom 2 to fix the per-block rates. By Axiom 2 (Axiom 2), the random rate α_i on block B_i is independent of the random rate $\alpha_{i'}$ on block $B_{i'}$ for $i \neq i'$. By Proposition 4.3.1, once the rates are realized, the rate-conditional channel matrix $Q_\ell^*(\alpha)$ for query ℓ has its row j determined by α_j alone: row j gives the conditional output distribution for an element in block B_j under the realized rate α_j. Write $(Q_\ell^*(\alpha))_{j_\ell, k_\ell}$ for the (j_ℓ, k_ℓ)-entry of this random matrix, the rate-conditional probability that query ℓ from block B_{j_ℓ} returns label k_ℓ given the rate α_{j_ℓ}. Distinct queries on the same block share the same realization of α_{j_ℓ} and hence the same row of $Q_\ell^*(\alpha)$; this row does not depend on what happened at other query positions. When all m queries share a single realization of the channel family (the common case for a stationary data structure), $Q_1^*(\alpha) = Q_2^*(\alpha) = \cdots = Q_m^*(\alpha) = Q^*(\alpha)$; the theorem's subscript ℓ flags per-query generality, not necessarily distinct matrices.

Step 3: apply Axiom 1 to factor the joint distribution. Conditional on the realized rates $\alpha = (\alpha_1, \ldots, \alpha_n)$, the outputs at positions $\ell = 1, \ldots, m$ are mutually independent, because each output is a deterministic function of the (conditionally independent) error event at x_ℓ. By Axiom 1 (Axiom

1), these error events are mutually independent given the rates and the objective set A. Therefore, conditional on the rates, the joint probability in (a) factors:

$$\begin{aligned}
&\mathrm{P}\left(\tilde{A}(x_1) = k_1, \ldots, \tilde{A}(x_m) = k_m | \alpha, x_\ell \in B_{j_\ell}\right) \\
&\quad = \prod_{\ell=1}^{m} \mathrm{P}\left(\tilde{A}(x_\ell) = k_\ell | \alpha_{j_\ell},\, x_\ell \in B_{j_\ell}\right) \\
&\quad = \prod_{\ell=1}^{m} (Q_\ell^*(\alpha))_{j_\ell, k_\ell}.
\end{aligned}$$

Step 4: read off the conditional matrix. The expression $\prod_{\ell=1}^{m}(Q_\ell^*)_{j_\ell,k_\ell}$ derived in Step 3 is, by definition, the $(j_1 \cdots j_m,\, k_1 \cdots k_m)$-entry of the Kronecker product $Q_1^*(\alpha) \otimes \cdots \otimes Q_m^*(\alpha)$. Since this holds for every index tuple, the rate-conditional joint matrix factors:

$$Q^{(m)*}(\alpha) \;=\; Q_1^*(\alpha) \otimes \cdots \otimes Q_m^*(\alpha). \tag{b}$$

This proves Theorem 4.4.1.

To obtain the unconditional version (Corollary 4.4.1.1) under the additional hypothesis that the queried blocks are pairwise distinct, take expectation over $\alpha = (\alpha_1, \ldots, \alpha_n)$:

$$\mathrm{E}\left[\prod_{\ell=1}^{m}(Q_\ell^*)_{j_\ell,k_\ell}\right] \;=\; \prod_{\ell=1}^{m}\mathrm{E}\left[(Q_\ell^*)_{j_\ell,k_\ell}\right] \;=\; \prod_{\ell=1}^{m}(Q_\ell)_{j_\ell,k_\ell}.$$

The first equality uses Axiom 2: when $j_1, \ldots, j_m$ are pairwise distinct, each factor depends on a distinct rate α_{j_ℓ}, and the cross-block independence guaranteed by Axiom 2 makes the factors mutually independent random variables. (When two queries hit the same block, this equality fails; see the second scope note above.) □

Honesty note. The parametric parsimony observation underlying this theorem appears as Remark 3.4 in the companion paper [Tow26a], which notes that Axiom 1 forces the joint confusion matrix to factor as a Kronecker product. The companion paper states this remark in a regime where rates are either deterministic or implicitly conditioned upon; under that reading the remark is correct, and the conditional form of Theorem 4.4.1 above makes the conditioning explicit. Axiom 2's role in this textbook is not to

extend the conditional factorization to the unconditional case in general. That is impossible without further hypotheses, since same-block queries on random rates produce a Jensen gap $\mathrm{E}\left[\alpha_j^2\right] > (\mathrm{E}\left[\alpha_j\right])^2$. Axiom 2 instead extends the unconditional factorization to the special case of distinct-block queries (Corollary 4.4.1.1) and, more importantly, certifies the rate-stability that lets us speak of "the" channel matrix Q_ℓ for each query position. The constructive proof above and the explicit conditional/unconditional decomposition are new contributions of this textbook.

Corollary: the parameter count drop

Corollary 4.4.1.2 (Parameter reduction, conditional form)**.** *Under Axioms 1 and 2, the rate-conditional joint confusion matrix $Q^{(m)*}(\alpha)$ of an n-th order Bernoulli predicate queried m times requires at most nm parameters to specify, rather than the $n(2^m - 1)$ parameters required by a general (unconstrained) m-query channel. For the order-2 Boolean predicate this reduces to $2m$ parameters (one per query per block) rather than $2(2^m - 1)$. The same count applies to the unconditional matrix $Q^{(m)}$ when rates are realized as constants (the deterministic-rates regime of all worked examples in this book) or when no block is queried twice (Corollary 4.4.1.1).*

Proof. We count the free parameters in $Q^{(m)*}(\alpha) = Q_1^*(\alpha) \otimes \cdots \otimes Q_m^*(\alpha)$.

Constrained case. Under the axioms, $Q^{(m)*}(\alpha)$ factors as $Q_1^*(\alpha) \otimes \cdots \otimes Q_m^*(\alpha)$ (Theorem 4.4.1), so each per-query matrix $Q_\ell^*(\alpha)$ contributes its own parameters independently. Each $Q_\ell^*(\alpha)$ is an $n \times 2$ stochastic matrix: rows are indexed by the n source blocks, and columns by the two binary output labels. Each row sums to 1, leaving 1 free parameter per row. With n rows, each $Q_\ell^*(\alpha)$ has n free parameters. By Axiom 2 the per-block rates are independent across queries, so the m matrices share no parameters. Across m queries: nm free parameters in total.

In the order-2 Boolean case ($n = 2$), the two blocks are positives and negatives, parametrized by (τ, ε) per query, or equivalently by (ω, ε). That is 2 parameters per query, for $2m$ parameters in total across m queries.

General case. The full block-tuple matrix has n^m rows and 2^m columns; specifying it freely would require $n^m(2^m - 1)$ parameters. The more compact count uses the per-source-block convention introduced above: without the Kronecker structure but still working source-block by source-block, each of the n source blocks contributes one (unconstrained) conditional distribution over 2^m joint output patterns, requiring $2^m - 1$ free parameters after

normalization. Total under this convention: $n(2^m - 1)$. This is the comparison point used here and in Section 4.3, because it reflects the reduction attributable to the axioms alone rather than conflating it with the block-tuple explosion.

Transfer to the unconditional matrix. When per-block rates are deterministic constants (so $Q^*_\ell(\alpha) = Q_\ell$ identically), the conditional and unconditional matrices coincide and the nm count applies to $Q^{(m)}$ directly. When rates are random but the queried blocks are pairwise distinct, Corollary 4.4.1.1 gives $Q^{(m)} = Q_1 \otimes \cdots \otimes Q_m$ with each Q_ℓ contributing n free parameters, again totalling nm. Both regimes cover every worked example in Part I.

Ratio.

$$\frac{n(2^m - 1)}{nm} \;=\; \frac{2^m - 1}{m},$$

which grows exponentially in m. At $m = 50$ and $n = 2$: general case $2(2^{50} - 1) \approx 2.25 \times 10^{15}$; constrained case $2 \times 50 = 100$. □

The corollary matches the closing summary of Section 4.3: the order-2 case requires $2m$ parameters under the axioms versus $2(2^m - 1)$ in the general case; the n-th order case requires nm versus $n(2^m - 1)$. The Kronecker structure, forced by the two axioms, is what produces this reduction.

Running thread: the BSC composed m times

The binary symmetric channel (BSC) introduced in **??** is the simplest non-trivial instance of the Kronecker theorem. A BSC with crossover probability ε has channel matrix

$$Q \;=\; \begin{pmatrix} 1-\varepsilon & \varepsilon \\ \varepsilon & 1-\varepsilon \end{pmatrix}.$$

Since the BSC is symmetric, ε is both the false-positive rate (an element outside A is reported as inside) and the false-negative rate (an element inside A is reported as outside). This is the symmetric special case of the order-2 Bernoulli model: a single crossover parameter ε suffices because the channel obeys FPR = FNR = ε (the symmetry constraint), and the two rows $[1-\varepsilon,\, \varepsilon]$ and $[\varepsilon,\, 1-\varepsilon]$ are reversals of each other. The model order is 2 because there are two partition blocks (B_1 = positives, B_2 = negatives); the symmetry constraint reduces the two free parameters of the general order-2 model to one.

Now suppose we run m independent queries, each through the same BSC. We work in the deterministic-rates regime announced in the second

scope note above, where $Q^*_\ell = Q$ for every query and the conditional matrix of Theorem 4.4.1 coincides with the unconditional one. The joint confusion matrix is then

$$Q^{(m)} \;=\; \underbrace{Q \otimes Q \otimes \cdots \otimes Q}_{m \text{ copies}} \;=\; Q^{\otimes m}.$$

Explicit computation for $m = 2$. Two queries, each through a BSC with crossover ε. The joint input lives in $\{B_1, B_1\} \cup \{B_1, B_2\} \cup \{B_2, B_1\} \cup \{B_2, B_2\}$, where we write B_1 for "in A" and B_2 for "not in A". The joint output lives in $\{(+,+), (+,-), (-,+), (-,-)\}$, where $+$ means "reported in A" and $-$ means "reported not in A".

By the Kronecker product definition, $Q \otimes Q$ is the 4×4 matrix formed by scaling each entry of Q by the full matrix Q:

$$Q^{\otimes 2} \;=\; Q \otimes Q \;=\; \begin{pmatrix} (1-\varepsilon)Q & \varepsilon Q \\ \varepsilon Q & (1-\varepsilon)Q \end{pmatrix}.$$

Expanding each block gives

$$Q^{\otimes 2} \;=\; \begin{pmatrix} (1-\varepsilon)^2 & (1-\varepsilon)\varepsilon & \varepsilon(1-\varepsilon) & \varepsilon^2 \\ (1-\varepsilon)\varepsilon & (1-\varepsilon)^2 & \varepsilon^2 & \varepsilon(1-\varepsilon) \\ \varepsilon(1-\varepsilon) & \varepsilon^2 & (1-\varepsilon)^2 & (1-\varepsilon)\varepsilon \\ \varepsilon^2 & \varepsilon(1-\varepsilon) & (1-\varepsilon)\varepsilon & (1-\varepsilon)^2 \end{pmatrix}.$$

Reading the matrix: the $(1,1)$-entry is the probability that two elements from block B_1 (both positives) are both reported positive, namely $(1-\varepsilon)^2$. The $(1,4)$-entry is the probability that two positives are both reported negative, namely ε^2. The $(2,2)$-entry—one positive, one negative; both reported positive—is $(1-\varepsilon)\varepsilon$: the positive is correctly reported positive $(1-\varepsilon)$ and the negative is falsely reported positive (ε).

For a numerical example, take $\varepsilon = 0.05$.

Joint input blocks	Joint output probabilities			
	$(+,+)$	$(+,-)$	$(-,+)$	$(-,-)$
(B_1, B_1)	0.9025	0.0475	0.0475	0.0025
(B_1, B_2)	0.0475	0.9025	0.0025	0.0475
(B_2, B_1)	0.0475	0.0025	0.9025	0.0475
(B_2, B_2)	0.0025	0.0475	0.0475	0.9025

The BSC thread that opened in **??** reaches its formal resolution here: m independent queries through the same BSC produce a joint channel whose confusion matrix is the m-fold Kronecker power $Q^{\otimes m}$. Each entry of $Q^{\otimes m}$

is a product of m individual BSC transition probabilities, each equal to ε or $1-\varepsilon$ according to whether that query position produced an error. The Kronecker structure is not a coincidence: it is exactly what Axioms 1 and 2 force.

A 2-query worked verification

We now verify Theorem 4.4.1 with a general order-2 channel to show that the BSC example is not special.

Let the predicate $\tilde{A}$ have false-positive rate ε and false-negative rate ω (so true-positive rate $\tau = 1-\omega$ and true-negative rate $\eta = 1-\varepsilon$). The per-query channel matrix is

$$Q \;=\; \begin{pmatrix} \tau & \omega \\ \varepsilon & \eta \end{pmatrix} \;=\; \begin{pmatrix} 1-\omega & \omega \\ \varepsilon & 1-\varepsilon \end{pmatrix},$$

where row 1 corresponds to positives ($x \in A$, block B_1) and row 2 corresponds to negatives ($x \notin A$, block B_2).

Suppose we query two elements $x_1 \in B_{j_1}$ and $x_2 \in B_{j_2}$ and observe outputs k_1 and k_2 respectively. For concreteness, take both queries to the same channel (both $Q_1 = Q_2 = Q$, since the rates are stationary by Proposition 4.3.1, which relies on both Axioms 1 and 2). As in the BSC thread, we work in the deterministic-rates regime, where the conditional matrix of Theorem 4.4.1 coincides with the unconditional one.

Theorem 4.4.1 predicts:

$$Q^{(2)} \;=\; Q \otimes Q.$$

Let us compute $Q \otimes Q$ directly and then verify each entry against its direct probability calculation.

The Kronecker product $Q \otimes Q$.

$$Q \otimes Q = \begin{pmatrix} \tau \cdot Q & \omega \cdot Q \\ \varepsilon \cdot Q & \eta \cdot Q \end{pmatrix}$$

$$= \begin{pmatrix} \tau^2 & \tau\omega & \omega\tau & \omega^2 \\ \tau\varepsilon & \tau\eta & \omega\varepsilon & \omega\eta \\ \varepsilon\tau & \varepsilon\omega & \eta\tau & \eta\omega \\ \varepsilon^2 & \varepsilon\eta & \eta\varepsilon & \eta^2 \end{pmatrix}.$$

Row order is $(B_1B_1, B_1B_2, B_2B_1, B_2B_2)$ and column order is $(++, +-, -+, --)$.

Direct probability calculation. We verify two entries by direct calculation.

Entry $(B_1B_1, ++)$: two positives, both reported positive. By Axiom 1, the events are independent:

$$\mathrm{P}\left(\tilde{A}(x_1) = +, \tilde{A}(x_2) = + | x_1 \in B_1, x_2 \in B_1\right) \;=\; \tau \cdot \tau \;=\; \tau^2.$$

This matches $(Q \otimes Q)_{11} = \tau^2$. Axiom 1 licenses the product; Axiom 2 ensures each τ factor is the same stable value.

Entry $(B_2B_1, -+)$: negative then positive, first reported negative (correct), second reported positive (correct).

$$\mathrm{P}\left(\tilde{A}(x_1) = -, \tilde{A}(x_2) = + | x_1 \in B_2, x_2 \in B_1\right) \;=\; \eta \cdot \tau.$$

This matches $(Q \otimes Q)_{33} = \eta\tau$ (row 3, column 3; note column 3 is the $(-,+)$ output pattern).

Entry $(B_1B_2, +-)$: positive then negative, both correctly reported. Block $B_1 = \{\top\}$ (positive), output $+$ is a true positive; block $B_2 = \{\bot\}$ (negative), output $-$ is a true negative.

$$\mathrm{P}\left(\tilde{A}(x_1) = +, \tilde{A}(x_2) = - | x_1 \in B_1, x_2 \in B_2\right) \;=\; \tau \cdot \eta.$$

This matches $(Q \otimes Q)_{22} = \tau\eta$. For the false-positive entry—output $(+,+)$ when blocks are (B_1, B_2)—we need column 1: $(Q \otimes Q)_{21} = \tau\varepsilon$, which equals $\mathrm{P}(+,+|B_1, B_2) = \tau \cdot \varepsilon$. Confirmed.

All entries are products of individual channel probabilities. This is exactly what Theorem 4.4.1 predicts: each row of $Q^{(2)}$ is the tensor product of the corresponding per-query rows, and the resulting matrix has 4 parameters (ε and ω per query, with both queries sharing the same rates by Axiom 2), not the $2(2^2 - 1) = 6$ that a general 2-query channel would need.

What the Kronecker structure buys for §??

The AND derivation in **??** computed the joint error rate of two noisy bits by treating the two error events as independent and multiplying probabilities. We can now see exactly which theorem that derivation was tacitly invoking.

The two-query joint channel for an AND gate is a special case of $Q^{(2)} = Q_1 \otimes Q_2$ where Q_1 and Q_2 are the per-input channel matrices for the two operands. The independence assumed in **??** was Axiom 1 (the two error events are independent) plus the stability of each operand's rates (Axiom 2).

The Kronecker factorization theorem makes this precise: the joint distribution of the two AND operands' channel outputs factors, and the AND gate's output distribution is a deterministic function of that joint distribution. Nothing more was needed than what the two axioms provide.

Closing: the formal type is next

The Kronecker theorem is the formal payoff for chapters 1–4's accumulated independence assumptions. We now have a complete characterization of the joint confusion matrix under the Bernoulli framework's two axioms: it is a Kronecker product of per-query channel matrices, one factor per query, with nm parameters total for an n-th order predicate on m queries.

In Section 4.5, we turn this machinery into a formal definition of the Bernoulli[bool] type—the object that chapters 1–3 have been pointing at. The Kronecker theorem is what makes that definition tractable: it tells us that the formal type is parametrized by finitely many numbers (two, for the Boolean case), not by an exponentially large family of joint distributions.

4.5 The Bernoulli[bool] Type, Formally

We have arrived at the formal definition. Four sections of scaffolding have built up to this moment: Section 4.1 explained why axioms matter, Section 4.2 stated that error events at distinct inputs are mutually independent, Section 4.3 stated that per-block error rates are mutually independent across blocks, and Section 4.4 proved that those two axioms together force the joint confusion matrix to factor as a Kronecker product. The formal definition is not an assertion; it is a synthesis of what the preceding chapters have earned.

The definition

Definition 4.5.1 (Bernoulli[bool])**.** *A* Bernoulli[bool] instance *is a triple* (X, Y, Q) *where:*

1. $X = \{\top, \bot\}$ *is the* input domain *(the Boolean universe), with* $B_1 = \{\top\}$ *and* $B_2 = \{\bot\}$ *as the two partition blocks.*
2. $Y = \{\top, \bot\}$ *is the* output domain *(the observed report).*
3. Q *is a* 2×2 *row-stochastic matrix*

$$Q = \begin{pmatrix} \tau & \omega \\ \varepsilon & \eta \end{pmatrix} = \begin{pmatrix} 1-\omega & \omega \\ \varepsilon & 1-\varepsilon \end{pmatrix},$$

where row 1 governs elements in $B_1 = \{\top\}$ (positives) and row 2 governs elements in $B_2 = \{\bot\}$ (negatives), with

- $\tau \in [0, 1]$*: the true-positive rate (*P (*reported* $\top$ | *true value* $\top$)*),*
- $\omega = 1 - \tau$*: the false-negative rate (*P (*reported* $\bot$ | *true value* $\top$)*),*
- $\varepsilon \in [0, 1]$*: the false-positive rate (*P (*reported* $\top$ | *true value* $\bot$)*),*
- $\eta = 1 - \varepsilon$*: the true-negative rate (*P (*reported* $\bot$ | *true value* $\bot$)*).*

The triple satisfies Axiom 1 (Axiom 1: error events at distinct inputs are mutually independent) and Axiom 2 (Axiom 2: per-block error rates ε and ω are conditionally independent of each other given the objective set).

Notation. *We write* Bernoulli[bool] *for the* type*: the collection of all triples (X, Y, Q) satisfying the definition above. An individual predicate or data structure is an* instance *of* Bernoulli[bool] *when its behavior matches Definition 4.5.1. The two parameters (ε, ω)—or equivalently (ε, τ)— uniquely identify the channel matrix Q within the type.*

Several remarks on what the definition does and does not say.

Scope. This definition is for the Boolean special case: the input and output domains are both $\{\top, \bot\}$, giving a 2×2 channel matrix with two free parameters. The Bernoulli framework extends to arbitrary algebraic types—Bernoulli[T] for general T—but that extension is Part III material and does not appear here. For now, all instances are Boolean.

The role of the two axioms. The definition invokes both axioms explicitly because both are load-bearing. Axiom 1 (Axiom 1) is what licenses the factored-product form of joint distributions: queries on distinct Boolean inputs produce independent error events. Without it, the channel matrix Q could not be read as a per-element description; correlation structure across inputs would require a far richer representation. Axiom 2 (Axiom 2) is what makes Q time-invariant: the false-positive rate ε and false-negative rate ω are jointly independent, so knowing one tells you nothing about the other, and neither varies with query history. Both axioms together are what makes Q a compact, two-parameter object sufficient to characterize an entire predicate's behavior—the content of Corollary 4.4.1.2.

The Q convention. Row 1 of Q corresponds to $B_1 = \{\top\}$ (positives); row 2 corresponds to $B_2 = \{\bot\}$ (negatives). Column 1 is the "report $\top$" output; column 2 is the "report $\bot$" output. This convention was introduced in Section 4.3 and used throughout Section 4.4; it is consistent across the entire book.

Biased-coin worked example

The biased-coin example has threaded through this book since **??**. We introduced it there as one of the three faces of a noisy bit—a coin with bias p for heads, where the observation itself may be noisy. In **??** we looked at the Boolean universe qualitatively. We deferred the full joint-probability treatment to this section. Here it lands.

Setup. Let the coin have bias p: heads ($\top$) with probability p, tails ($\bot$) with probability $1-p$. The observation is noisy: it reports correctly for positives with probability $\tau = 1-\omega$ and correctly for negatives with probability $\eta = 1-\varepsilon$.

Example 1 (Biased coin as a Bernoulli[bool] **instance)** *Let $U = \{\top, \bot\}$, with $A = \{\top\}$ the objective set (the event "coin lands heads") and $A^c = \{\bot\}$ the complement (the event "coin lands tails"). Define:*

- *Coin bias: $p = \mathrm{P}(\textit{true value} = \top)$.*
- *Channel: false-positive rate ε (probability of reporting heads when the coin actually landed tails), false-negative rate ω (probability of reporting tails when the coin actually landed heads).*

The joint PMF over all four (true value, observed report) pairs is:[1]

Outcome	*True value*	*Observed report*	*Probability*
True positive (TP)	$\top$	$\top$	$p(1-\omega) = p\,\tau$
False negative (FN)	$\top$	$\bot$	$p\,\omega$
False positive (FP)	$\bot$	$\top$	$(1-p)\,\varepsilon$
True negative (TN)	$\bot$	$\bot$	$(1-p)(1-\varepsilon) = (1-p)\,\eta$

Sanity check:

$$\begin{aligned} p\,\tau + p\,\omega + (1-p)\,\varepsilon + (1-p)\,\eta &= p(\tau+\omega) + (1-p)(\varepsilon+\eta) \\ &= p\cdot 1 + (1-p)\cdot 1 \;=\; 1. \quad \checkmark \end{aligned}$$

The channel matrix is:

$$Q \;=\; \begin{pmatrix} \tau & \omega \\ \varepsilon & \eta \end{pmatrix} \;=\; \begin{pmatrix} 1-\omega & \omega \\ \varepsilon & 1-\varepsilon \end{pmatrix}.$$

[1] This formulation differs from the companion paper's Example 3.5, which gives the PMF over the realized approximate-set value rather than the joint distribution over true-vs-observed pairs; the two are equivalent but suit different purposes.

The triple $(\{\top, \bot\}, \{\top, \bot\}, Q)$ *is a* Bernoulli[bool] *instance whenever* ε *and* ω *are conditionally independent given the coin's objective set* $A = \{\top\}$*, which holds when the observation mechanism operates independently on each side of the partition. The prior* p *over the input is not part of the* Bernoulli[bool] *type itself—the type characterizes the* channel*, not the* source*—but* p *enters when we compute the joint or marginal distribution over outcomes.*

For a concrete instance, take $p = 0.6$*,* $\varepsilon = 0.05$*,* $\omega = 0.10$ *(so* $\tau = 0.90$*,* $\eta = 0.95$*). The four outcome probabilities are:*

Outcome	*Formula*	*Value*
TP	0.6×0.90	0.540
FN	0.6×0.10	0.060
FP	0.4×0.05	0.020
TN	0.4×0.95	0.380
Total		1.000

The channel matrix for this instance is:

$$Q \;=\; \begin{pmatrix} 0.90 & 0.10 \\ 0.05 & 0.95 \end{pmatrix}.$$

Axiom 1 is vacuously satisfied for this single-query example: there is only one input element, so "independence across distinct inputs" has no content. Axiom 1's operational role appears only with multi-query joint distributions (the setup of Theorem 4.4.1), where it justifies the factored joint matrix. Here we are checking that the per-instance type structure does not violate Axiom 1, not that we exercise it.

Axiom 2 governs the joint distribution over random per-block rates; this concrete instance fixes a single realization $(\varepsilon, \omega) = (0.05, 0.10)$*. The choice of these two values without coupling between them is consistent with Axiom 2's cross-block independence. The full content of Axiom 2 emerges when we consider rates as random variables drawn from a joint distribution; see Proposition 4.3.1 and the closing summary of Section 4.3.*

This triple is a well-formed Bernoulli[bool] *instance.*

The biased-coin example is the simplest general (asymmetric) instance of the type. It has one free parameter in the source (p) and two in the channel (ε, ω). The source prior p controls how often each block is queried; the channel parameters control how reliably each query is reported. The formal Bernoulli[bool] type isolates the channel: it says exactly what can go wrong (and how often) for each kind of input, independently of how often that input arises.

Chapters 1 through 3 as instances

The formal definition gives us a vocabulary for what we have been doing all along. We can now look back at every example in chapters 1–3 and say, precisely, what type each object was.

Chapter 1: the noisy bit (??). The three faces introduced in **??**—biased coin, binary symmetric channel, classifier verdict—are all Bernoulli[bool] instances. The biased coin appears above as Example 1. The binary symmetric channel has $\varepsilon = \omega$ (symmetric noise), so its channel matrix is $Q = \left[\begin{smallmatrix} 1-\varepsilon & \varepsilon \\ \varepsilon & 1-\varepsilon \end{smallmatrix}\right]$, a special case of the general Q above. The classifier verdict has ε and ω determined by the classifier's decision boundary and the data distribution—arbitrary in general, but still a Bernoulli[bool] instance whenever Axioms 1 and 2 hold.

In **??** we walked through the Boolean universe qualitatively, deferring the full PMF treatment to here. That deferral has now paid off: Example 1 gives the PMF table, and Definition 4.5.1 gives the formal type.

Chapter 2: binary channels (??, ??). The binary symmetric channel of **??** is exactly Bernoulli[bool] with Q symmetric: $\varepsilon = \omega$ reduces the two free parameters of the general Q to one. The formal Bernoulli[bool] type is strictly more general; the BSC lives as a one-parameter subfamily.

The asymmetric binary channel of **??** is Bernoulli[bool] with Q general: ε and ω may differ freely. The Z-channel (one-sided error) is the extreme case $\omega = 0$ or $\varepsilon = 0$. The Bloom filter, which reports false positives but never false negatives, is the $\omega = 0$ instance: a row-1 channel that always reports $\top$ correctly and a row-2 channel that reports $\top$ with probability $\varepsilon > 0$.

Chapter 3: AND, OR, NOT under noise (??, ??, ??). The gate-level operations of chapter 3 are operations on Bernoulli[bool] instances that produce new Bernoulli[bool] instances.

AND: if $\tilde{b}_1$ has rates $(\varepsilon_1, \omega_1)$ and $\tilde{b}_2$ has rates $(\varepsilon_2, \omega_2)$, the AND of two noisy bits derived in **??** produces a new Bernoulli[bool] instance with closed-form ε_{AND} and ω_{AND}. The derivation in **??** implicitly invoked Axiom 1 (the two error events are independent) and Axiom 2 (the rates are stable across the joint computation). Now that both axioms are named, we can say it precisely: AND is a binary operation Bernoulli[bool] $\times$ Bernoulli[bool] $\to$ Bernoulli[bool].

OR: by De Morgan's duality, the OR of two Bernoulli[bool] instances is also a Bernoulli[bool] instance. Its rates can be derived from AND via De Morgan or computed directly from the truth table of the noisy

OR (**??**); either way, the operation has type signature Bernoulli[bool] × Bernoulli[bool] → Bernoulli[bool]. AND and OR together generate the rest of Boolean algebra under noise.

NOT: complement swaps the block labels, exchanging ε and ω. Let $\varepsilon' = \omega$ and $\omega' = \varepsilon$ denote the effective rates of the complemented instance. The result is a Bernoulli[bool] instance with

$$Q_{\text{NOT}} = \begin{pmatrix} 1-\omega' & \omega' \\ \varepsilon' & 1-\varepsilon' \end{pmatrix} = \begin{pmatrix} 1-\varepsilon & \varepsilon \\ \omega & 1-\omega \end{pmatrix}.$$

Here the new instance's effective false-positive rate is the original ω and its effective false-negative rate is the original ε. NOT is a unary operation Bernoulli[bool] → Bernoulli[bool].

The composition theorem of **??**—that error rates compose multiplicatively in series—is the special case of Theorem 4.4.1 (the Kronecker factorization) applied to chained Bernoulli[bool] instances. Each stage of the chain is one factor in the Kronecker product; the total joint distribution is the m-fold tensor $Q_1 \otimes \cdots \otimes Q_m$. This is not a coincidence: both rest on the independence structure that the two axioms supply. The composition theorem (matrix product, serial composition) and the Kronecker theorem (tensor product, parallel composition) are sister consequences of the same axiomatic foundation operating in different geometries.

What the formal definition earned

A pedagogy that opens with a formal definition and then works out examples is a different kind of book. This book chose the other order: examples first, patterns second, formal definition last. The benefit is that the definition arrives already supported by intuition—the reader knows what Bernoulli[bool] means before reading the words.

The cost of that ordering is the distance the reader must carry intuition before it is formalized. In **??**, we described the Boolean universe qualitatively and said the full treatment would come later. That deferred a cognitive debt. This section retires it.

Chapters 1–3 built the intuition. Section 4.2 and Section 4.3 named the independence assumptions those chapters were silently invoking. Section 4.4 showed what those assumptions buy in terms of factorization and parameter reduction. Definition 4.5.1 collects all of that into a single, precise statement.

This is the atom-outward trajectory: intuition-first, formalism at the end of Part I rather than the beginning. The formal definition was not asserted; it was discovered, step by step, over four chapters. The reader who has

followed those chapters does not need to be persuaded that the definition is the right one. They have seen it emerge.

In Section 4.6, we look ahead at where this definition leads.

4.6 What Comes Next

We have been building toward a single formal object—Bernoulli[bool] as defined in Section 4.5—and we have arrived. This is a good moment to take stock of where we stand and where Part II will take us.

What Part I delivered

Chapter 4 has completed Part I's formal work. The two axioms are now stated with precision (Sections 4.2 and 4.3), the Kronecker factorization theorem shows what those axioms license (Theorem 4.4.1), and Definition 4.5.1 gives Bernoulli[bool] a rigorous identity as the triple (X, Y, Q) under those axioms. Every example in chapters 1 through 3—the biased coin, the binary symmetric channel, the classifier verdict, AND and OR and NOT under noise, the k-round Miller-Rabin bound—is now recognizable as an instance or an operation on instances of this type. The definition was not asserted; it was earned.

Five Core Principles: status after Part I

The Bernoulli framework rests on five organizing principles. At the close of Part I, three are fully in view and two remain ahead.

Two Axioms **[Part I, complete**] Axiom 1 (element-wise independence: Section 4.2) and Axiom 2 (conditional independence of block error rates: Section 4.3) are both stated formally and proved to jointly imply the Kronecker factorization. They are the load-bearing assumptions throughout the rest of the book.

Composition Closure **[Part I, complete**] The composition theorem of **??** showed that chaining Bernoulli[bool] instances produces a new Bernoulli[bool] instance. AND, OR, and NOT are all closure-preserving operations. The Kronecker theorem of Theorem 4.4.1 is the tensor-product face of the same closure property: whether we compose instances serially (matrix product) or view them in parallel (tensor product), the type is closed. The Boolean universe is self-contained under the operations we care about.

Boolean Thread [**Part I, complete**] Part I chose $\mathsf{Bool} = \{\top, \bot\}$ as the input and output domain and stayed there. Two elements, two partition blocks, two error rates, a 2×2 channel matrix, two free parameters after Corollary 4.4.1.2 applies—the whole Part I story has Boolean structure at every level. This is not a limitation; it is the atom's defining feature. The Boolean thread runs from the biased coin in chapter 1 to the formal definition in chapter 4 without deviation. Part II lifts it to sets; Part III lifts it to maps.

Parametric Parsimony [**visible in Part I, completed in Part II**] The corollary to the Kronecker theorem (Corollary 4.4.1.2) showed the parameter count dropping from $n(2^m - 1)$ to $2m$ under the two axioms. For the Boolean case, that reduction is already dramatic. The full principle—why $2m$ parameters are not merely convenient but are, in an information-theoretic sense, the right number—requires the entropy-based lower bounds developed in Part II, chapter 8. The seed has been planted; the fruit arrives later.

Space-Accuracy Duality [**Part II, chapter 8**] This principle—that there is a fundamental trade-off between the space a Bernoulli structure consumes and the error rates it can achieve—has not yet appeared. Part I did not need it because we were analyzing a fixed Bernoulli[bool] instance, not asking how small we can make one while meeting an accuracy target. Space-Accuracy Duality is the central organizing principle of Part II. It arrives in chapter 8.

From bits to sets

Part I gave us the atom. Part II asks: what happens when you build a set membership predicate from atoms?

The question is less abstract than it sounds. A set S over a universe U induces, for each element $u \in U$, a Boolean query: is $u \in S$? An *approximate* set replaces exact membership with a noisy channel: the predicate may report $u \in S$ when it should report $u \notin S$ (a false positive) or the reverse (a false negative). The per-element error events are—under Axiom 1—mutually independent. Each element's query is a Bernoulli[bool] instance. The approximate set as a whole is a structured collection of those instances, and its behavior is determined by how those instances are built and indexed.

The canonical first construction is the Bloom filter. A Bloom filter maintains a bit array of m bits and a family of k hash functions; it answers set-membership queries with zero false negatives and a tunable false-positive

rate. Understanding the Bloom filter from the Bernoulli perspective—as a mechanism for manufacturing Bernoulli[bool] instances with $\omega = 0$ and a controlled ε—is the project of Chapter 5. The Bloom filter is not a special case we study because it is famous; it is the cleanest first construction because it makes the $\omega = 0$ boundary and the Space-Accuracy Duality both visible in one concrete object.

By the end of Chapter 5, the reader will see not just how the Bloom filter works but why its analysis is a consequence of the formal machinery we developed in Part I.

Part II then extends: chapter 6 studies how set operations (union, intersection, complement) propagate error rates; chapter 7 develops the classification measures that quantify approximate-set quality; and chapter 8 derives the information-theoretic lower bounds that make Space-Accuracy Duality precise. The Boolean thread does not disappear in Part II—it becomes the foundation on which more complex structures are built, one Bernoulli[bool] instance at a time.

Bibliographic Notes

The material in this chapter draws on three streams of literature.

The Kronecker product. The product that carries Leopold Kronecker's name appears in his 1866 work on bilinear forms, though its use in matrix algebra spread gradually through the following century. Kronecker himself was interested in the algebraic structure of determinants and quadratic forms; the representation-theoretic and probabilistic uses of the product came later. For a modern treatment of the Kronecker product's algebraic properties—including the mixed-product property, vectorization identities, and spectral structure—see Horn and Johnson [HJ91], whose Chapter 4 is the standard graduate reference. The formal theorem statement in Section 4.4 of this textbook—that under Axioms 1 and 2 the joint confusion matrix factors as a Kronecker product of per-query matrices—is a new contribution; see the honesty note at the end of Section 4.4 for a precise account of what is new and what descends from the companion papers.

Information-theoretic channels. Chapter 4's proof of the Kronecker factorization uses the language of channels and conditional distributions but does not develop information theory from scratch. Readers who want the theoretical background—binary symmetric channels, channel capacity,

and the source-channel coding theorem—should consult Cover and Thomas [CT06]. Chapters 7 and 8 of that work cover binary channels at the depth Part I uses; Chapter 10 covers channel capacity. We borrow the channel-matrix vocabulary and the notion of a memoryless channel (which is, in the Bernoulli setting, precisely what Axioms 1 and 2 enforce), but we do not reproduce Cover and Thomas's development here.

The Bernoulli papers. The two-axiom framework that organizes this chapter comes from Towell [Tow26a]. Axioms 1 and 2, as formally stated in Sections 4.2 and 4.3, are reformulations of the axioms in that paper; the textbook's contribution is to separate them into named sections, give each its own worked examples, and prove the Kronecker factorization theorem as an explicit consequence rather than treating it as a remark. The companion paper presents the factorization in a regime where rates are deterministic or implicitly conditioned upon; the textbook makes the conditional and unconditional forms explicit (Theorem 4.4.1 and Corollary 4.4.1.1), noting that the unconditional form requires either deterministic rates or pairwise-distinct queried blocks. The model-order analysis—the idea that the partition of the input domain into blocks is what governs the parameter count—appears in Towell [Tow26b], which extends the Bernoulli framework from Boolean sets to approximate values over arbitrary algebraic types. The order-2 Boolean case treated in this chapter is the simplest instance of that general theory.

This textbook reformulates and extends both papers. The formal theorem statement in Section 4.4, the full constructive proof, and the explicit connection between model order and the Kronecker product's parameter count are contributions of this text.

Exercises

1. ⋆**Axiom 1 in measure-theoretic notation.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the underlying probability space, and for each $x \in U$ define the error event $E_x = \{\omega \in \Omega : \tilde{A}(x)(\omega) \neq \mathbf{1}_A(x)\} \in \mathcal{F}$. Reformulate Axiom 1 (Axiom 1) using this notation: write out the independence condition both pairwise and for an arbitrary finite family $\{x_1, \ldots, x_k\} \subseteq U$. Does the condition depend on the specific probability measure $\mathbb{P}$, or only on the sigma-algebra $\mathcal{F}$?

2. ⋆**Axiom 2 as a constraint on Q.** Restate Axiom 2 (Axiom 2) as a property of the channel matrix Q. Specifically: in the order-2 Boolean case, Q has two rows. Axiom 2 declares the random quantities

governing those rows to be conditionally independent. Express this as a statement about the joint density $f(\varepsilon, \omega | A)$. What behavior does Axiom 2 *forbid* Q from exhibiting across a sequence of queries? (See Proposition 4.3.1 for the stability consequence derived in §4.3.)

3. ⋆**Computing a Kronecker product.** Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}.$$

Compute $A \otimes B$ by hand. Verify that $A \otimes B$ is a 4×4 matrix. Then compute $B \otimes A$ and check whether $A \otimes B = B \otimes A$. (*Hint:* the Kronecker product is not in general commutative; the two products will differ, but both will be 4×4.)

4. ⋆⋆**Proving the parameter-count corollary.** Prove Corollary 4.4.1.2 by direct counting.

   (a) Without any independence assumption, an n-th order model on m joint queries has $n(2^m - 1)$ free parameters. Explain where this count comes from.

   (b) Under Axioms 1 and 2, the Kronecker factorization (Theorem 4.4.1) forces the parameter count down to nm. Show why: how many free parameters does each per-query matrix Q_i contribute, and why do the m per-query matrices together total nm?

   (c) For the order-2 Boolean channel ($n = 2$), compute the ratio $n(2^m - 1)/(nm)$ for $m = 10$ and $m = 50$. Comment on the magnitude of the reduction.

5. ⋆⋆**Verifying the Kronecker theorem by hand for $m = 2$.** Use the asymmetric channel from the biased-coin worked example (Example 1) with $\varepsilon = 0.05$ and $\omega = 0.10$, so $Q = \left(\begin{smallmatrix} 0.90 & 0.10 \\ 0.05 & 0.95 \end{smallmatrix}\right)$.

   (a) Compute $Q \otimes Q$ explicitly. Write out all sixteen entries of the resulting 4×4 matrix.

   (b) For the two-query input pair $(\top, \bot)$, use direct probability to compute $\mathrm{P}\left(\tilde{A}(\top) = s_1,\ \tilde{A}(\bot) = s_2 | A\right)$ for all four outcome pairs $(s_1, s_2) \in \{\top, \bot\}^2$.

   (c) Verify that the entries from (b) match the corresponding row of $Q \otimes Q$ from (a), confirming Theorem 4.4.1 for $m = 2$.

6. ⋆⋆**A Markov channel violates Axiom 1.** The Markov-correlated channel in §4.3 couples the error rate on each query to the outcome of the previous query: if the previous query produced an error, the current query uses elevated false-positive rate $\varepsilon_1 > \varepsilon_0$. Consider two consecutive queries on distinct elements $x, y \in B_2 = \{\bot\}$, with base rate $\varepsilon_0 = 0.05$ and elevated rate $\varepsilon_1 = 0.30$.

   (a) Compute $\mathrm{P}(E_x = 1,\, E_y = 1)$ under the Markov channel, where the channel starts in the correct state.

   (b) Compute $\mathrm{P}(E_x = 1) \cdot \mathrm{P}(E_y = 1)$ under the same parameters.

   (c) Show that these two quantities differ, thereby exhibiting a violation of Axiom 1 (Axiom 1).

   (*Note:* §4.3 frames this example as an Axiom 2 failure; here we recast it explicitly as an Axiom 1 failure. Both framings are valid; the Markov coupling violates both axioms simultaneously.)

7. ⋆⋆**The implicit Axiom-2 invocation in §3.1's AND derivation.** Re-read the derivation of AND under noise in **??**. The derivation writes the joint probability that both inputs to an AND gate produce errors as a product of two individual error probabilities.

   (a) Identify the step where Axiom 2 (Axiom 2) is invoked implicitly. Quote or describe the exact expression where the factoring occurs.

   (b) Suppose Axiom 2 failed: the false-positive rate on the first input told you something about the false-positive rate on the second. Write an expression for the joint error probability in that case (in terms of the conditional density $f(\varepsilon_2|\varepsilon_1)$).

   (c) Would the closed-form AND output rate from §3.1 survive if Axiom 2 failed? Explain in one or two sentences.

8. ⋆⋆**Constructing a** Bernoulli[bool] **from a biased coin.** Following Definition 4.5.1 and Example 1, construct a Bernoulli[bool] instance for a biased coin with bias $p = 0.7$, false-positive rate $\varepsilon = 0.10$, and false-negative rate $\omega = 0.05$.

   (a) Write down the channel matrix Q and the joint PMF over all four (true value, observed report) pairs.

   (b) Verify Axiom 1 (Axiom 1) in the following sense: the Boolean universe has only two elements ($\top$ and $\bot$), so state what elementwise independence means in this degenerate case and confirm it holds by construction.

(c) Verify Axiom 2 (Axiom 2) in the following sense: state the joint density condition $f(\varepsilon, \omega|A) = f(\varepsilon|A)\, f(\omega|A)$ and explain why it is satisfied when the observation mechanism acts independently on each partition block.

9. ⋆⋆⋆**Counterexample: Axiom 1 does not imply Axiom 2.** Construct a probability model that satisfies Axiom 1 (Axiom 1: error events at distinct elements are mutually independent) but violates Axiom 2 (Axiom 2: per-block error rates are conditionally independent across blocks).

   (a) Describe the model explicitly: specify U, the partition blocks, and the joint distribution of error rates across blocks.

   (b) Verify that Axiom 1 holds: for any fixed realization of the block rates, the per-element error events are independent.

   (c) Show that Axiom 2 fails: the joint density of the block rates does not factor.

   (*Hint:* consider a two-block model where the rates ε and ω are drawn from a bivariate distribution with nonzero correlation, such as a Dirichlet or a copula, while the per-element draws given those rates remain independent.)

10. ⋆⋆⋆**Without Axiom 2, the channel-matrix view collapses.** Show that without Axiom 2 (Axiom 2), the joint distribution over m queries to an element in a fixed block B_i cannot in general be written as a product of m identical channel-matrix rows.

    (a) Give a concrete two-query example where the marginal per-query distribution is consistent with a channel matrix Q, yet the *joint* two-query distribution over outcomes for a single block B_i does not equal $(Q_{i,\cdot})^{\otimes 2}$. (You may use the Markov-correlated channel from §4.3 as your example.)

    (b) What mathematical object would replace the single matrix Q in a model without Axiom 2? Name or describe an appropriate generalization (e.g., a time-varying sequence of matrices, a hidden Markov model, or a conditional channel).

    (c) Explain in one paragraph why the Kronecker factorization theorem (Theorem 4.4.1) depends on Axiom 2 and not only on Axiom 1.

11. ⋆⋆⋆**Tightness of the parameter-count bound.** Corollary 4.4.1.2 states that under Axioms 1 and 2, an n-th order Bernoulli predicate on m queries requires nm free parameters. Argue that this bound is tight: there exist Bernoulli[bool] instances satisfying both axioms that *genuinely* require $2m$ parameters to specify (in the order-2 Boolean case).

    (a) Construct a family of Bernoulli[bool] instances parameterized by m independently chosen channel matrices $Q_1, Q_2, \ldots, Q_m$, each with its own $(\varepsilon_i, \omega_i)$. Verify that this family satisfies both axioms.

    (b) Show that no two distinct members of the family yield the same joint distribution over m queries, so distinct parameter tuples $(\varepsilon_1, \omega_1, \ldots, \varepsilon_m, \omega_m)$ produce distinct joint confusion matrices.

    (c) Conclude that the $2m$ parameters are not merely sufficient but necessary: no reparameterization can reduce the family to fewer than $2m$ free parameters without collapsing distinct members.

Part II

From Bits to Sets

Chapter 5

Approximate Sets and Predicates

Placeholder. Content drafted in Plan 2.

Chapter 6

Composition of Sets

Placeholder. Content drafted in Plan 2.

Chapter 7

Classification Measures

Placeholder. Content drafted in Plan 2.

Chapter 8

Information-Theoretic Bounds

Placeholder. Content drafted in Plan 2.

Part III

From Sets to Maps

Chapter 9

Approximate Functions and n-ary Channels

Placeholder. Content drafted in Plan 3.

Chapter 10

Sketches and Estimators

Placeholder. Content drafted in Plan 3.

Chapter 11

Randomized Algorithms

Placeholder. Content drafted in Plan 3.

Chapter 12

Approximate Relations

Placeholder. Content drafted in Plan 3.

Chapter 13

The Algebra of Approximate Types

Placeholder. Content drafted in Plan 3.

Part IV

Optimality and Construction

Chapter 14

Optimal Bernoulli Sets

Placeholder. Content drafted in Plan 4.

Chapter 15

Perfect Hash Filters

Placeholder. Content drafted in Plan 4.

Part V

The Boundary and Beyond

Chapter 16

Approximate but Not Bernoulli

Placeholder. Content drafted in Plan 4.

Chapter 17

Cipher Maps and Trapdoor Computing

Placeholder. Content drafted in Plan 4.

Chapter 18

Encrypted Search as Relaxed Trapdoor

Placeholder. Content drafted in Plan 4.

Appendix A

Probability Refresher

Placeholder. Content drafted in Plan 5.

Appendix B

The Five Core Principles

Placeholder. Cheat-sheet of the Two Axioms, Parametric Parsimony, Space-Accuracy Duality, Composition Closure, and the Boolean Thread. Content drafted in Plan 5.

Appendix C

Library Walkthrough

Placeholder. Tour of the `bernoulli` *Python package as a unified entry point. Content drafted in Plan 5.*

Appendix D

Selected Exercise Solutions

Placeholder. Roughly half of the end-of-chapter exercises are worked here; the rest are reader practice. Content drafted in Plan 5.